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Enhancing epidemic preparedness: a data-driven system for managing respiratory infections

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Abstract

Background Effective epidemic preparedness is critical for minimizing the health and societal impacts of viral respiratory infections. This study details the development of a data-driven early warning system (EWS) designed to improve outbreak detection and response utilizing the data integration and visualization capabilities of Microsoft Power BI.

Methods This research utilized a structured three-phase approach to design a respiratory infections (RIs) management dashboard. Phase 1, focused on identifying critical variables through literature reviews and expert interviews. In Phase 2, Microsoft Power BI was employed for dashboard development, integrating data from diverse sources. Phase 3 involved usability testing with health professionals who evaluated navigation, data accuracy, decision-support features, providing feedback to enhance visualization clarity and filtering capabilities.

Results Key data categories include individual-level variables, such as age, symptoms, and vaccination records, alongside population-level metrics like infection rates and regional vaccination coverage enabling functionalities such as identifying high-risk individuals, tracking infection dynamics, and optimizing resource allocation. The dashboard, developed using Power BI visualizes epidemiological trends, intervention outcomes, and resource utilization. A relational database schema ensures efficient data retrieval, facilitating comprehensive analysis.

Conclusion The prototype EWS represents a scalable and integrative framework aimed at enhancing public health applications, particularly in the context of respiratory infections. By incorporating data from diverse health sectors, the system offers decision-makers access to critical epidemiological indicators, supporting early outbreak detection and improved epidemic management. Its potential to unify health institutions underscores its value in fostering a more cohesive and effective approach to epidemic preparedness. Nevertheless, while the system demonstrates significant promise, further evaluation in real-world settings is essential to determine its practical impact on public health outcomes and its ability to mitigate health crises.

Keywords Epidemic preparedness, Data-Driven, Early warning system (EWS), Respiratory infections (RIs), Outbreak detection, Microsoft Power BI

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Introduction

Respiratory infections, caused by a range of pathogens and characterized by symptoms such as fever, cough, sore throat, and myalgia [1, 2] pose a significant public health burden. Their potential for rapid and widespread transmission can result in outbreaks, increased hospitalizations, and substantial mortality [3, 4, 5, 6, 7, 8, 9]. The recent COVID-19 pandemic further highlighted the critical need for real-time data systems and rapid diagnostics to inform effective decision-making during infectious disease outbreaks underscoring the vulnerability of global health security to emerging pathogens [10, 11]. The substantial impact of RIs is exemplified by seasonal influenza epidemics in the United States, which are estimated to cause 41 million illnesses, 710,000 hospitalizations, and 21 million medical visits annually [12, 13]. To effectively manage the complexities of infectious disease surveillance, advanced technologies such as electronic dashboards and business intelligence (BI) tools have become indispensable [14, 15]. These systems offer real-time data visualization, sophisticated analytics, and actionable insights, empowering decision-makers to respond swiftly and effectively during public health emergencies [16, 17, 18, 19, 20, 21, 22]. A key component of these systems is the incorporation of early warning mechanisms capable of detecting anomalies and monitoring outbreak trends [23]. Statistical process control charts, such as Cumulative Sums (CUSUM) effective in detecting small, sustained shifts in data, while Exponentially Weighted Moving Averages (EWMA) are more sensitive to recent changes [24, 25]. These methods are well-suited for analyzing time-sensitive healthcare data, enabling early detection of subtle deviations from expected trends, facilitating timely corrective actions, and supporting continuous monitoring of key performance indicators, thus allowing healthcare providers to proactively manage potential outbreaks [26, 27].

Golestan province in Iran has faced significant challenges, including a severe spring flood in 2019 [28] and the subsequent COVID-19 pandemic which strained its healthcare system and highlighted its vulnerability to large-scale emergencies. Historical events, such as the Yurt Mine explosion [29], further emphasize the region's susceptibility to crises that overload healthcare infrastructure. This study introduces a prototype early warning system for managing respiratory infections, developed using Microsoft Power BI. The system integrates dynamic visualizations, predictive modeling, and trend analysis to provide actionable insights for public health professionals. Key goals include integrating diverse data streams for real-time disease monitoring, employing predictive models to forecast outbreaks, and offering a user-friendly interface for decision-makers in both high- and low-resource settings. Central to the

initiative is a managerial dashboard leveraging business intelligence tools to enhance decision-making capabilities. Target users include public health officials, hospital administrators, epidemiologists, and disaster management authorities. The platform supports tracking epidemiological trends, assessing healthcare capacity, predicting caseloads for resource optimization, and preparing for surges in emergencies. To address data overload, the dashboard incorporates Situation Awareness Oriented Design (SAOD) [30] principles, aligning visualizations with cognitive processes to facilitate swift interpretation of complex datasets. This approach enhances situational awareness, enabling stakeholders to identify critical trends and implement timely interventions during public health crises. By combining advanced analytics with intuitive design, the system aims to strengthen preparedness and response strategies for respiratory infection management.

Material and method

This study employed a three stages methodology to develop and implement a comprehensive respiratory infection management dashboard, ensuring a rigorous and replicable process.

Phase 1: Identification of key variables

To establish the foundational dataset, a combined approach of a literature review of existing studies and dashboards, alongside semi-structured interviews with experts in public health, IT, health management, and epidemiology. A systematic review was conducted across databases such as PubMed, Scopus, Web of Science, and Google Scholar to identify a standardized minimum dataset for designing a Power BI-based dashboard for managing viral respiratory infections (VRIs). Using MeSH terms and expert consultation, keywords such as “*electronic health record, HER, database management system, DBMS, information management system, hospital information system, data management system, early warning system, influenza like illness, VRI, COVID 19, outbreak, dashboard*” and terms related to respiratory infections were refined, and search syntax was developed, with data extracted following PRISMA guidelines. We screened titles, abstracts, and full texts to gather relevant data. The inclusion/ exclusion criteria included: original papers, review articles, peer-reviewed studies, and English language articles specifically related to the minimum dataset and dashboards for ILI were included. Letters to the editor, conference papers, theses, brief articles, and blog reports were excluded.

A qualitative content analysis approach was employed to gather expert insights on essential design elements, focusing on features, attributes, and data requirements. Fifteen participants with at least five years of experience

in epidemiology, public health, IT, or data science were purposively sampled until data saturation was achieved. Each interview lasted between 45 and 60 min and was audio-recorded with participant consent. Transcripts were created from the recordings, and field notes were taken to capture additional insights and contextual nuances. Semi-structured interviews explored key dashboard features, navigation components, and real-time data integration needs. Transcripts were analyzed through open, axial, and selective coding to identify themes such as user interface design, data harmonization, and decision support requirements. Credibility was ensured via member checking, peer debriefing, and an audit trail. The findings provide a robust foundation for developing a user-friendly, data-driven Power BI dashboard optimized for real-time public health decision-making in the context of RIs. The variables encompassed critical aspects of infectious disease monitoring, including demographic, clinical, laboratory, geographic, and resource allocation data. Thematic analysis was used to synthesize qualitative data from the interviews, grouping variables into distinct categories and themes. A consensus-building workshop with healthcare stakeholders was then conducted to validate the relevance and applicability of these variables and ensure alignment with operational needs.

Phase 2: Dashboard Development process

Data integration and ETL process The dashboard was developed using Microsoft Power BI, chosen for its robust data integration capabilities, interactive visualization tools, and ability to handle large datasets efficiently. This platform allowed for seamless integration of data from multiple sources and the creation of user-friendly dashboards tailored to the needs of public health professionals.

NAB Primary healthcare data, providing demographic details and socio-economic factors.

Hospital Information Systems (HIS) Clinical data, including symptoms, diagnoses, hospitalizations, and discharge outcomes.

Laboratory Information Systems (LIS) Test results, positivity rates, and diagnostic trends.

Ministry of Health Portals for Influenza Surveillance System Data on ILI cases, including test results and vaccination records.

Medical Crisis Management Command (MCMC) Emergency data such as ICU admissions and ventilator usage (Table 1).

Prior to loading into Power BI, data was consolidated and pre-processed in Microsoft Excel. This included rigorous data cleaning to remove duplicate records (identified using unique identifiers such as patient national ID, name, and surname) and address inconsistencies. The data then underwent an Extract, Transform, and Load (ETL) process: [1] Extraction: Retrieving raw data; [2] Transformation: Cleaning, standardizing, and calculating key metrics (prevalence, incidence, hospitalization rates, and test positivity); [3] Loading: Importing the refined data into Power BI (Fig. 1).

Dashboard design and feature development The dashboard’s structure and features were designed to provide actionable insights for decision-makers. Power BI components such as DAX (Data Analysis Expressions), Power Query, and Power Map were utilized to create an interactive and user-friendly interface. Key sections of the dashboard include.

Demographic analysis Visualizing population characteristics, including age, gender, and socio-economic factors, to identify vulnerable groups.

Trend analysis Monitoring case counts, hospitalization metrics, and vaccination rates over time.

Predictive tools Employing control charts (CUSUM and EWMA) to forecast potential outbreaks and assess intervention effectiveness.

Geospatial mapping Highlighting hotspots and resource distribution using dynamic maps.

Table 1 Data sources and key variables extracted from health subsystems

Subsystem	Data Type	Key Variables	Purpose/Use in Dashboard
NAB	Primary HealthCare Data	Demographic details, socio-economic factors, geographic settlement	Identify vulnerable populations by demographic and socioeconomic factors.
HIS	Clinical Data	Symptoms, diagnosis, hospitalizations	Monitor ILI trends and identify case clusters.
LIS	Laboratory Data	Test results, test positivity rate	Track infection rates and evaluate testing coverage.
MCMC	EMS Emergency Data	ICU admissions, ventilator use	Assess ICU utilization, track severe cases for surge response planning.
MoH Portal	ILI Data	Symptoms, diagnosis, test results	Serve as the primary source of ILI case information for outbreak detection.

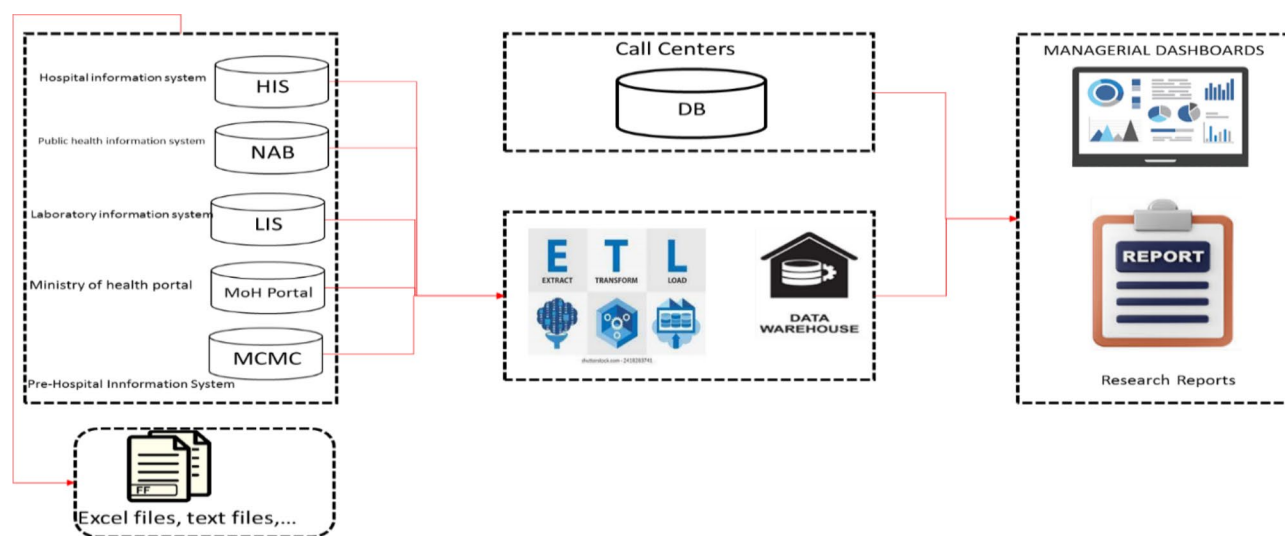


Fig. 1 The ETL process for Respiratory Infection Surveillance

The dashboard design prioritized usability, incorporating intuitive navigation menus and interactive visualizations to facilitate rapid decision-making. Performance optimization techniques, such as data aggregation and query reduction, were applied to ensure real-time responsiveness.

Phase 3: Validation and testing

Upon completion of the system design and dashboard creation, validation of the dashboard was conducted using real-world data from the aforementioned healthcare subsystems. Usability testing was conducted with 15 public health professionals with experience in infectious disease surveillance and management. Participants were selected based on their roles and expertise in epidemiology, public health administration, data analysis. The usability testing assessed the dashboard based on three key criteria: [1] Ease of Navigation: the accessibility and intuitiveness of the user interface; [2] Accuracy: the reliability of the data and insights provided; and [3] Decision-Support Functionality: the effectiveness of the dashboard in informing resource allocation and outbreak management. Specific metrics used in the usability testing included task completion time, error rates. Feedback from the validation process was used to refine the dashboard, addressing issues such as visualization clarity and data filtering options. The current system can now be accessed through the provided link within the intranet of Golestan University of Medical Sciences (<http://bi.goums.ac.ir/>).

Prototype evaluation

The prototype underwent retrospective data analysis to evaluate its reliability and practical application. Historical datasets were employed to assess the system's

performance in accurately detecting trends and anomalies. This approach allowed for a systematic examination of its accuracy and responsiveness under simulated conditions. However, the prototype phase was not without limitations, as it primarily relied on retrospective data rather than real-time inputs, which may affect the generalizability of the findings.

Results

Through literature review and expert interview we extracted various data types, including demographic information (e.g., population size, age distribution, income levels), clinical metrics (e.g., confirmed cases, recoveries, fatalities), hospital-related parameters (e.g., bed capacity, admission rates), exposure data (e.g., contact tracing, travel history), and risk factors (e.g., pre-existing conditions) for managerial dashboard reaction. Geographic coordinates are incorporated to enable geo-spatial mapping of cases and healthcare facilities, while data from primary healthcare providers and laboratories enhances the understanding of testing capacity (Table 2).

After finalizing the dataset, a user-friendly interface was developed in Power BI. Several important steps were undertaken, including the careful selection of relevant variables such as age, gender, residential location, risk factors, vaccination history, diagnostic test results, treatment history, and exposure to confirmed cases. It is designed to be located in the office of the President of the Golestan University of Medical Sciences (GOUMS) and will serve as a tool for monitoring trends in infectious diseases during epidemics and outbreaks.

Core sections and extracted elements The dashboard was structured into several innovative sections based on user needs and extracted datasets. The developed dash-

Table 2 Overview of individual and population-level variables in data management and early warning systems (EWS) for RIs management dashboard

Category	Variables	Level	Purpose in EWS
Demographics	"Age", "Gender", "Residential address", "Socio-economic status (income, education)"	Individual	Identify vulnerable individuals based on demographic factors.
	"Population age distribution", "Gender distribution by region", and "Socio-economic disparities"	Population	Assess infection patterns across different population segments.
Clinical Data	"Symptoms (fever, cough, shortness of breath)", "Comorbidities (diabetes, hypertension)"	Individual	Monitor health risks and severity at the individual level.
	"Incidence rates", "Prevalence rates", "Mortality trends", "Recovery rates"	Population	Monitor and predict infection dynamics over time.
Laboratory Data	"Test results (PCR, antigen tests)", "Turnaround time for diagnostics"	Individual	Assess diagnostic accuracy and detect infections early.
	"Testing facility density", "Test positivity rates by Region", "Diagnostic turnaround trends"	Population	Ensure equitable access to diagnostic services.
Vaccination Status	"Immunization record (vaccine type, doses completed)", and "Vaccination date"	Individual	Ensure individuals are up-to-date on vaccination.
	"Regional vaccination rates", "Urban vs. rural coverage", and "Age-specific vaccination statistics"	Population	Identify areas with immunization gaps to guide vaccination campaigns.
Resource Utilization	"ICU admission status", "Ventilator usage", "Length of hospital stay"	Individual	Track resource use for personalized care.
	"ICU bed availability", "Hospital capacity usage", "Staffing levels by region"	Population	Optimize allocation of healthcare resources during outbreaks.
Outcome Data	"Recovery status", "Mortality indicators", and "Time to recovery"	Individual	Measure treatment success and patient outcomes.
	"Mortality trends segmented by age and region", "Recovery rate comparisons across regions"	Population	Evaluate the overall effectiveness of public health interventions.
Travel and Exposure	"Travel exposure details", "Event attendance"	Individual	Identify exposure risks and possible sources of infection.
	"Travel-related exposure hotspots", "High-risk public event zones"	Population	Monitor travel-related outbreaks and community-level risks.
Risk Factors	"Occupation exposure", "Smoking status", "Household contact with confirmed cases".	Individual	Highlight personal risk factors contributing to infection or severity.
	"Vulnerability mapping (elderly populations, comorbid groups)", "Behavioral risks (hesitancy)"	Population	Identify population-wide risk factors to target interventions.
Geospatial Mapping	"Residential address visualization", "Case-specific location data."	Individual	Monitor individual mobility and exposure risk in specific locations.
	"Regional hotspots", "Resource distribution maps (ICU, ventilators)", "Travel-related zones"	Population	Visualize high-risk areas for targeted interventions.
Predictive Analytics	"Control chart trends for individual infection risk"	Individual	Identify early warning signs of worsening conditions in specific cases.
	"EWMA forecasts", "CUSUM signals", "Seasonal infection trends"	Population	Enable proactive responses to predicted surges in cases.

board offers valuable insights, and the relational database schema demonstrates the underlying data structure, which supports efficient data retrieval and analysis. This is essential for integrating various datasets and generating data-driven insights (Fig. 2).

Figure 3 Integrated data model in Microsoft Power BI illustrating the relational structure of data from multiple sources used to develop the respiratory infection early warning system. The diagram shows the various tables (LabServiceAdmC, NAB, vacsan_view3, MergeHIS-LAB, Sheet1, and PersonalTotal) and the established relationships between them (indicated by connecting lines). This integrated model enables data flow and analysis across

different data domains within the Power BI dashboard, facilitating comprehensive surveillance and reporting.

Figure 4. Comparison of rolling average and Exponentially Weighted Moving Average (EWMA) for analyzing time series data of respiratory infection cases across different locations in Golestan province. The dashboard view displays individual plots for each location (e.g., Gorgan, Gonbad, Azadshahr, etc.), with the rolling average and EWMA overlaid to highlight trends, seasonal variations, and potential outbreak signals. This visualization facilitates comparison of epidemiological patterns across different regions.

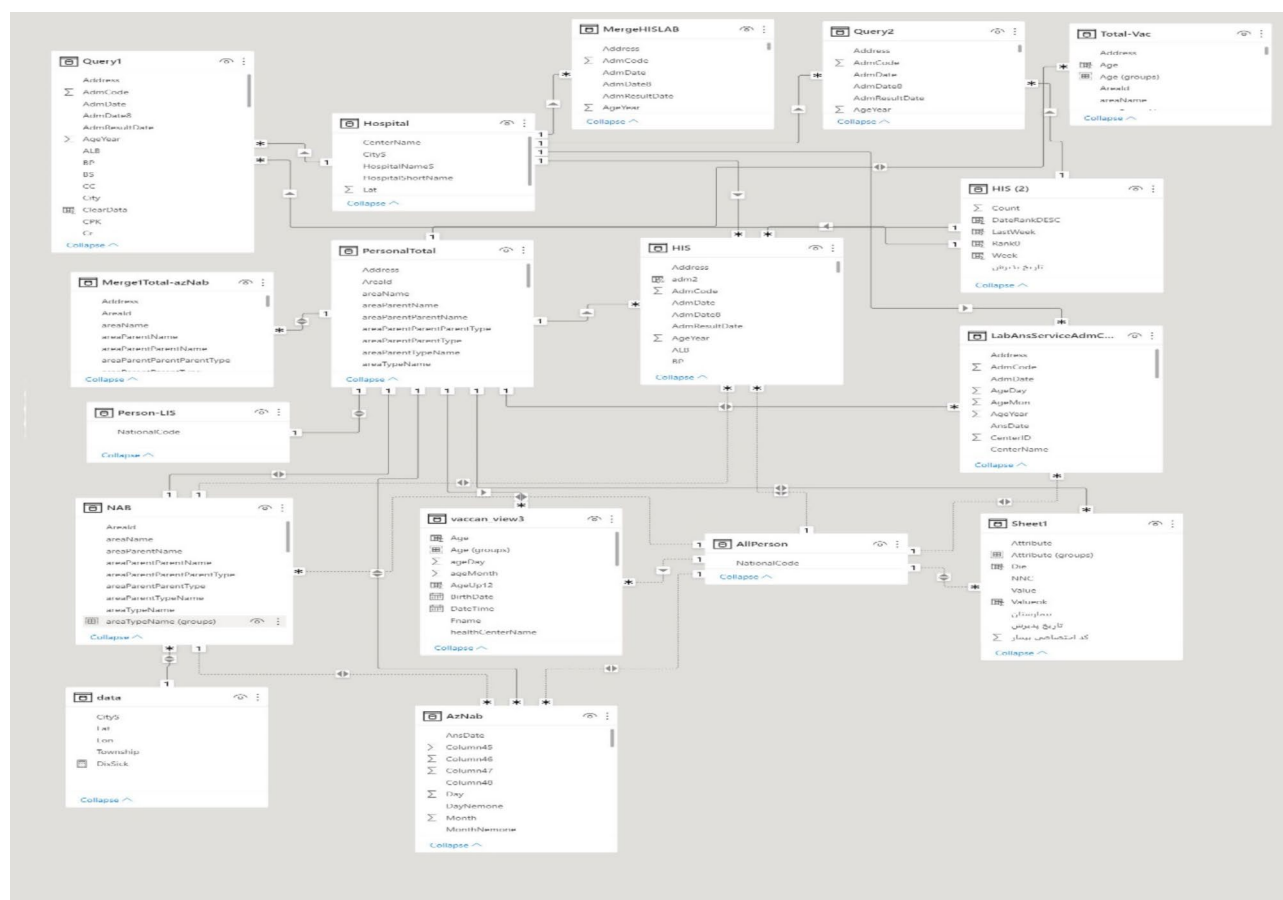


Fig. 2 Data model The diagram shows the data tables derived from multiple sources, including the public health registry system (NAB), Hospital Information Systems (HIS), Laboratory Information Systems (LIS), Ministry of Health Portal (MoH Portal), and Medical Crisis Management Command

Predictive analytics panel Comparison of raw data and control chart methods for two-week RI case prediction. The left panel presents the raw time series data of RI cases, showing fluctuations over time. The right panel illustrates the application of CUSUM and EWMA control charts to the same dataset. The CUSUM chart (represented by highlights cumulative deviations from the mean, while the EWMA chart emphasizes recent changes in the data. This comparison demonstrates the potential of control charts for early detection of changes in RI case trends and improved short-term forecasting (Fig. 5).

Epidemiological insights: Trend analysis module Key performance indicators displayed on the respiratory infection management dashboard. The dashboard provides a comprehensive overview of the epidemiological situation, including: [1] the total number of patient visits (66,961); [2] the total number of unique patients (52,213); [3] the mortality rate relative to the population (0.32%); [4] a time series plot illustrating the daily trend of hospitalizations, showing fluctuations and peaks over time; and [5] a pie chart depicting the gender distribution of hospitalized patients. This integrated view allows for rapid

assessment of the outbreak's impact and informs public health decision-making (Fig. 6).

Geospatial mapping Spatial distribution of respiratory infection indicators across Golestan province as visualized in the management dashboard. The top two maps show the distribution of cases over different cities of Golestan province (incidence rate). The bottom two maps depict the location of different healthcare facilities from health houses at rural areas to managerial centers at urban zones. This visualization facilitates the identification of hotspots and informs targeted interventions. (Fig. 7).

Discussion

This study described the development and implementation of a novel, integrated EWS for managing respiratory infections, leveraging the data integration and visualization capabilities of Microsoft Power BI. This dashboard integrates control charts (CUSUM and EWMA), geospatial mapping, and predictive analytics to provide a potentially powerful tool for enhancing epidemiological surveillance and informing public health decision-making.

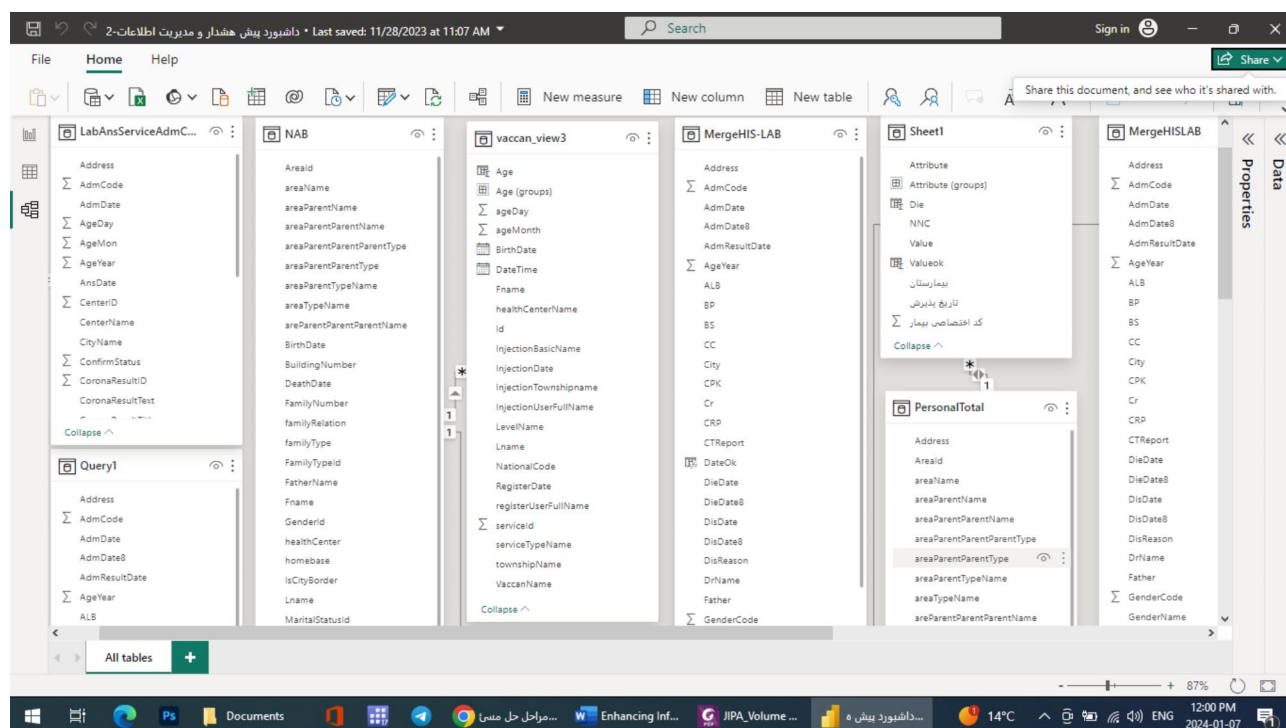


Fig. 3 Power BI data model illustrating table relationships for the respiratory infection early warning system



Fig. 4 Time series plots of respiratory infection cases with rolling average and EWMA overlay

The development of a standardized minimum dataset (MDS) was a crucial first step. This process, informed by both a systematic literature review and expert interviews, ensured the inclusion of key variables relevant to

RI surveillance and management. This approach aligns with the work of Zarei et al. (2021) [31], who also emphasized the importance of establishing an MDS for disease dashboards. However, this study extended this approach

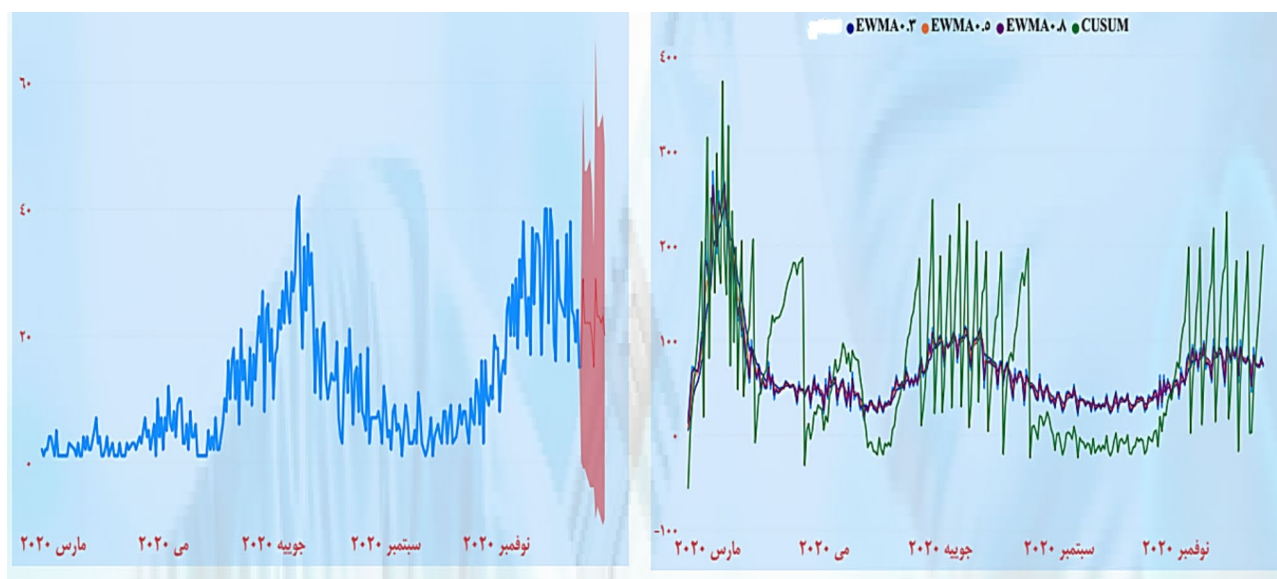


Fig. 5 Trend analysis and anomaly detection using cumulative Sum (CUSUM) and exponentially weighted moving average (EWMA) control charts

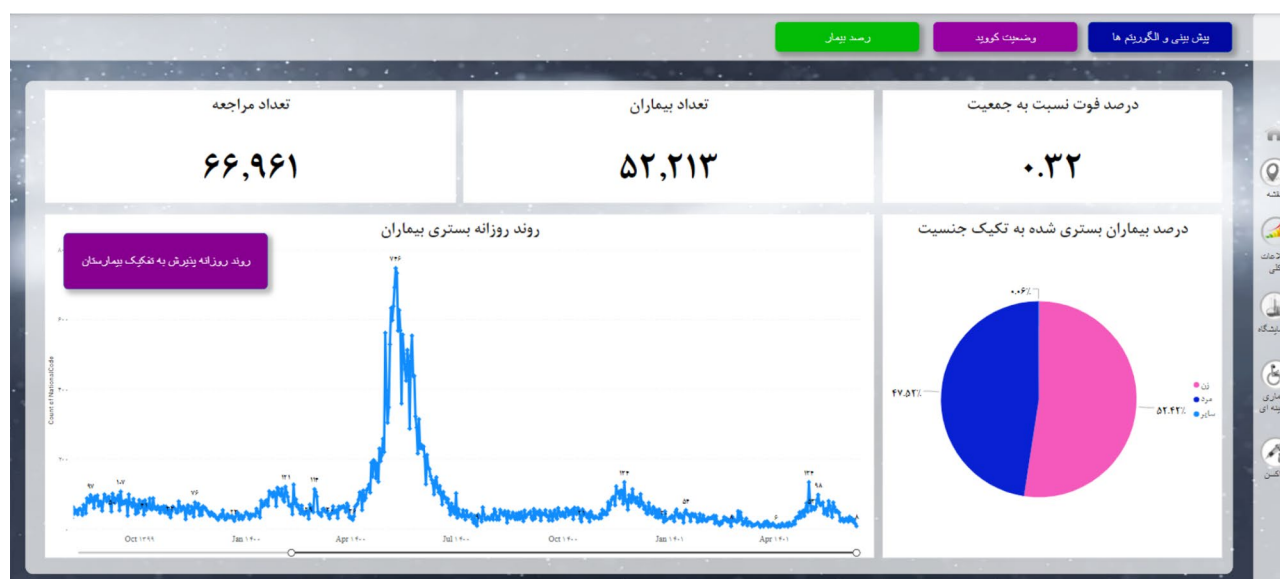


Fig. 6 Dashboard interface showing key metrics related to patient visits, total patients, mortality rate, daily hospitalizations, and gender distribution of hospitalized patients

by incorporating geospatial data and focusing specifically on RIs. The challenge of data integration from disparate sources, as highlighted by a 2021 study evaluating Iranian COVID-19 registries [31, 32], further underscores the need for such integrated systems. This dashboard addresses this challenge by using ETL processes.

The application of control charts (CUSUM and EWMA) for early outbreak detection is a key feature of this EWS. Previous studies have demonstrated the effectiveness of these methods in detecting outbreaks of other infectious diseases, such as measles (Karami et al., 2012, 2013) [33, 34] and influenza (Solgi et al., 2018) [35]. This

study contributes to this body of literature by applying these methods to ILI data in the context of Golestan province. While Karami et al. (2012) found Holt-Winters exponential smoothing effective for removing seasonal patterns [36], this study focused on using CUSUM and EWMA for real-time outbreak detection based on daily case counts.

The implementation of Power BI has enabled the creation of an interactive, user-friendly platform for dynamic data visualization and analysis, aligning with prior research on leveraging technology for public health monitoring. This aligns with the findings of Beltran et

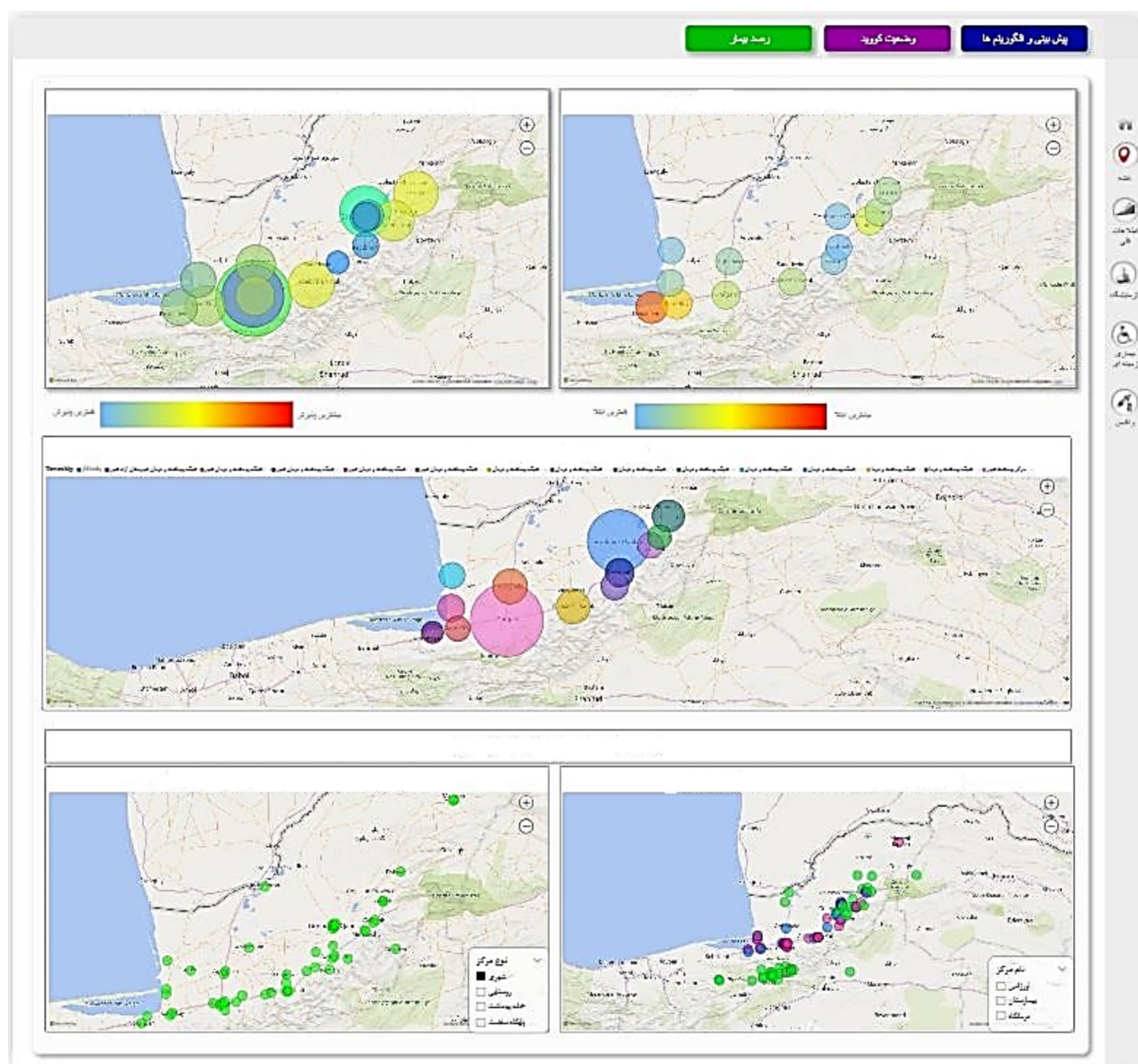


Fig. 7 Geospatial distribution of respiratory infection indicators

al. (2021) [37], who developed the CONVIDa system for COVID-19 data collection and analysis, and Lubogo et al. (2022) [38], who implemented an Electronic-Based Early Warning Alert and Response Network (EWARN) in Somalia. Similarly, studies by Hosseini et al. (2018) [39], Wang et al. (2014) [40], and Jonnalagadda [41] highlight the utility of advanced visualization tools in public health crises. Power BI's real-time charts facilitate trend identification and temporal comparisons of epidemics, as demonstrated by Dixon et al. [42] who created population-level dashboards for COVID-19 surveillance. This study integrates control charts for influenza prediction into a comprehensive dashboard featuring geospatial mapping and vaccine distribution tracking, enhancing

decision-making capabilities. Inspired by Tina Damalas' 2024 work [43] on addressing vaccine inequities in Pima County, this tool not only monitors vaccination status but also visualizes vaccine distribution and administration schedules across locations. Such advancements underscore the importance of data-driven approaches in improving situational awareness and supporting equitable public health interventions during epidemics.

Implications for Public Health Practice and Research

This dashboard has the potential to enhance public health responses to RIs and other infectious diseases, particularly in settings with limited resources. The real-time alert system, triggered by pre-defined thresholds,

can prompt timely public health interventions. The integration of critical care capacity data facilitates more efficient resource allocation during outbreaks. By tracking vaccination coverage and identifying at-risk populations, the dashboard can also support targeted vaccination campaigns. These capabilities could be particularly valuable in LMICs (Low- and Middle-Income Countries) and other regions facing similar challenges to our region.

Future research should focus on evaluating the real-world impact of the dashboard on public health outcomes. This could involve conducting prospective studies to assess its effectiveness in detecting outbreaks earlier, improving resource allocation, and reducing morbidity and mortality. Further research could also explore user adoption and satisfaction with the dashboard, as well as its adaptability to other infectious diseases and settings.

Conclusion

This study introduces a prototype Early Warning System (EWS) aimed at enhancing epidemic preparedness and response for respiratory infections. Developed using Microsoft Power BI, the system integrates data visualization, predictive modeling, and geospatial mapping into an accessible platform designed for public health decision-makers. By consolidating data from multiple sources, the dashboard facilitates improved outbreak detection, resource allocation, and identification of high-risk areas. Usability testing demonstrated the system's potential to support timely and informed decision-making in public health contexts.

Limitations

This study presents some limitations that merit attention. The dashboard's effectiveness heavily depends on the quality and completeness of its data, with inaccuracies or gaps potentially undermining the reliability of its visualizations and predictions. Usability testing involved a small sample of 15 participants, limiting the generalizability of findings; future research should incorporate larger, more diverse populations. Validation was based on retrospective data, which, while useful, does not replicate real-time outbreak scenarios, necessitating live-environment assessments in subsequent studies. Technological reliance poses challenges in areas with limited internet access or infrastructure, emphasizing the importance of ongoing maintenance to sustain its functionality. Prototype testing revealed additional constraints, such as reliance on simulated data, absence of automated integration with national health systems, and untested scalability in varying resource settings. Predictive models, though validated on historical data, require further evaluation under dynamic outbreak conditions to establish their practical accuracy. Addressing these limitations in future work is

essential to improve the system's robustness, scalability, and adaptability across diverse public health contexts.

Abbreviations

EWS	Early warning system
RIs	Respiratory infections
COVID-19	Coronavirus disease of 2019
EWMA	Exponentially Weighted Moving Averages
CUSUM	Cumulative Sum
SAOD	Situation Awareness Oriented Design
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
ETL	Extract, transform, load
NAB	(in Persian = Nezame Etelaat Behdasht (in English = public health information system))
HIS	Hospital Information System
LIS	Laboratory Information System
MCMC	Medical Crisis Management Command
MoH	Ministry of Health
DAX	Data Analysis Expressions
GOUMS	Golestan University of Medical Sciences
LMICs	Low- and Middle-Income Countries

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12879-025-10528-y>.

Supplementary Material 1

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Author contributions

KJ and MS designed the research. MS collected the data. KJ, MS, MH, and MK participated in data analysis. All authors have prepared the primary manuscript and then finalized it.

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Data availability

The current system can now be accessed through the provided link within the intranet of Golestan University of Medical Sciences (<http://bi.goums.ac.ir/>).

Declarations

Ethics approval and consent to participate

This study adhered to the principles of the Helsinki Declaration and received ethical approval from Shahid Beheshti University of Medical Sciences (IR.SBMU.PHNS.REC.1402.003). Informed consent was obtained from all participants involved in the usability testing phase. No personal or sensitive data were used during system testing.

Consent for publication

Not applicable. (Since no identifying personal, clinical, or image data are presented in the manuscript).

Competing interests

The authors declare no competing interests.

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References

- Kittikraisak W, Wongraee T, Punjasamanvong S, Piyaraj P, Vachiraphan A, Yoocharoen P, et al. Influenza-like Symptom Incidence, Illness-Associated expenses, and Economic Impact among Healthcare Personnel in Thailand: a prospective Observational Cohort Study (2020–2021). *Ann Work Expo Health*. 2023;67(3):330–44.
- Tao J, Gao H, Zhu S, Yang L, He D. Influenza versus COVID-19 cases among influenza-like illness patients in travelers from Wuhan to Hong Kong in January 2020. *Int J Infect Dis*. 2020;101:323–5.
- Gefenaita G, Pistol A, Popescu R, Popovici O, Ciurea D, Dolk C, et al. Estimating burden of influenza-associated influenza-like illness and severe acute respiratory infection at public healthcare facilities in Romania during the 2011/12–2015/16 influenza seasons. *Influenza Other Respir Viruses*. 2018;12(1):183–92.
- Akhtar Z, Ghosh P, Bhuiyan M, Sturm-Ramirez K, Rahman M, Howlader M, et al. Influenza-associated respiratory illness among five cohorts of pregnant women and their young infants (0–6 months), Bangladesh, 2013–2017. *Influenza Other Respir Viruses*. 2023;17(8):e13175.
- Faye MN, Barry MA, Jallow MM, Wade SF, Mendy MP, Sy S et al. Epidemiology of Non-SARS-CoV2 Human coronaviruses (HCoVs) in people presenting with influenza-like illness (ILI) or severe Acute Respiratory infections (SARI) in Senegal from 2012 to 2020. *Viruses*. 2022;15(1).
- Potdar V, Vijay N, Mukhopadhyay L, Aggarwal N, Bhardwaj SD, Choudhary ML, et al. Pan-india influenza-like illness (ILI) and severe acute respiratory infection (SARI) surveillance: epidemiological, clinical and genomic analysis. *Front Public Health*. 2023;11:1218292.
- Sechan F, Edridge AWD, van Rijswijk J, Jebbink MF, Deijs M, Bakker M, et al. Influenza-like illness symptoms due to endemic human coronavirus reinfections are not influenced by the length of the interval separating reinfections. *Microbiol Spectr*. 2024;12(3):e0391223.
- Shedura VJ, Hussein AK, Nyanga SK, Kamori D, McHau GJ. Evaluation of the influenza-like illness sentinel surveillance system: a national perspective in Tanzania from January to December 2019. *PLoS ONE*. 2023;18(3):e0283043.
- Macias AE, McElhaney JE, Chaves SS, Nealon J, Nunes MC, Samson SI, et al. The disease burden of influenza beyond respiratory illness. *Vaccine*. 2021;39(Suppl 1):A6–14.
- Pakdaman A, Karajizadeh M, Nasiri M, Sharifian R. Business Intelligence Dashboard in Healthcare: lesson learned. *Health Management and Information Science*. 2022;Health Management Inform Sci;9(3).
- Aba Oud M, Almuqrin M. On the early detecting of the COVID-19 outbreak. *J Infect Dev Ctries*. 2021;15(11):1625–9.
- Chan EYY, Dubois C, Fong AHY, Shaw R, Chatterjee R, Dabral A, et al. Reflection of challenges and opportunities within the covid-19 pandemic to include biological hazards into drr planning. *Int J Environ Res Public Health*. 2021;18(4):1614.
- Sayah Mofazali A, Jahangiri K. Towards a customized foresight model on "Disaster risk management" in developing countries. *Foresight* 2018; 20(5): 467–87. <https://doi.org/10.1108/FS-01-2018-0002>
- JINPON P, JAROENSUTASINEE M, KJWJoS JAROENSUTASINEE. Business intelligence and its applications in the public healthcare system. *J Sci Technol*. 2011;8(2):97–110.
- Trincanato E, Vagnoni EJJoHO M. Business intelligence and the leverage of information in healthcare organizations from a managerial perspective: a systematic literature review and research agenda. *J Health Organ Manag*. 2024;38(3):305–30.
- Gu X, Watson C, Agrawal U, Whitaker H, Elson W, Anand S, et al. Post-pandemic development of sentinel surveillance of respiratory disease, in the context of the WHO mosaic framework: protocol for the English primary care network 2023–2024 (Preprint). *JMIR Public Health and Surveillance*; 2024.
- Ramalingam S, Subramanian M, Reddy AS, Tarakaramu N, Khan MI, Abdullaev S, et al. Exploring business intelligence applications in the healthcare industry: a comprehensive analysis. *Egypt Inf J*. 2024;25:100438.
- Ridgway JP, Lauderdale D, Thisted R, Robicsek A. Electronic syndromic surveillance for Influenza-Like Illness Across Treatment settings. *Infect Control Hosp Epidemiol*. 2017;38(4):393–8.
- Lubogo M, Karanja MJ, Mdodo R, Elnossery S, Osman AA, Abdi A, et al. Evaluation of the electronic early warning and Response Network (EWARN) system in Somalia, 2017–2020. *Confl Health*. 2022;16(1):18.
- Mohindra NA, Garcia SF, Kircher S, Barnard C, Perry LM, Lyleroeher M, et al. Development of an electronic health record-integrated patient-reported outcome-based shared decision-making dashboard in oncology. *JAMIA Open*. 2024;7(3):ooae056.
- Linder JA, Jung E, Housman D, Eskin MS, Schnipper JL, Middleton B et al. The Acute Respiratory Infection Quality Dashboard: a performance measurement reporting tool in an electronic health record. *AMIA Annu Symp Proc*. 2007:1035.
- Jaradat Z, Al-Dmour A, Alshurafat H, Al-Hazaima H, Al Shbail MO. Factors influencing business intelligence adoption: evidence from Jordan. *J Decis Syst*. 2024;33(2):242–62.
- Ren H, Li Y, Huang T. Anomaly Detection models for SARS-CoV-2 Surveillance based on genome k-mers. *Microorganisms*. 2023;11(11).
- Bernardo CO, Gonzalez-Chica DA, Chilver M, Stocks N. Influenza-like illness in Australia: a comparison of general practice surveillance system with electronic medical records. *Influenza Other Respir Viruses*. 2020;14(6):605–9.
- Suman G, Prajapati DJJM, Engineering Q. Control chart applications in healthcare: a literature review. *Int J Metrol Qual Eng*. 2018;9:5.
- Steiner SH, Grant K, Coory M, Kelly HA. Detecting the start of an influenza outbreak using exponentially weighted moving average charts. *BMC Med Inf Decis Mak*. 2010;10(1):1–8.
- Iqbal J, Noor-Ul-Amin M, Khan I, AlQahtani SA, Yasmeen U, Ahmad B. A novel bayesian Max-EWMA control chart for jointly monitoring the process mean and variance: an application to hard bake process. *Sci Rep*. 2023;13(1):21224.
- Sarani M, Safi-Keykaleh M, Safarpour H, Honarvar MR, Kalteh E, Jahangiri K. Tractor Ambulance: an Initiative of Emergency Medical Services Response to the Flood of Golestan, Iran in 2019. *Disaster Med Public Health Prep*. 2022;16(4):1695–7.
- Sarani M, Honarvar M, Sahebi A, Safi-Keykaleh M, Nateghinia S, Jahangiri K. Challenges facing the health system in responding to mining accidents: the case of the zemestan-yurt mine explosion in Iran (2017). *J Occup Health Epidemiol*. 2021;10(3):204–8.
- Azad FT, Candan KS, Kapkiç A, Li M-L, Liu H, Mandal P, et al. (Vision Paper) a vision for Spatio-Causal Situation Awareness, forecasting, and planning. *ACM Trans Spat Algorithms Syst*. 2024;10(2):1–42.
- Zarei J, Badavi M, Karandish M, Haddadzadeh Shoushtari M, Dastoorpoor M, Yousefi F, et al. A study to design minimum data set of COVID-19 registry system. *BMC Infect Dis*. 2021;21(1):1–13.
- Rampisheh Z, Kameli ME, Zarei J, Barzaki AV, Meraji M, Mohammadi A. Developing a national minimum data set for hospital information systems in the Islamic Republic of Iran Mise Au point d'un ensemble de données minimum Au Niveau national pour les systèmes d'information hospitaliers en République islamique d'Iran. *East Mediterr Health J*. 2020;26(4):400–9.
- Karami M, Soori H, Mehrabi Y, Haghdooost A, Gouya M. Evaluating the performance of the exponentially weighted moving average in measles outbreak detection using real data testing. *irje* 2012. 8(3):22–32. <http://irje.tums.ac.ir/article-1-5017-en.html>
- Karami M, Soori H, Mehrabi Y, Haghdooost A, Gouya M, Esmailnasab N. Evaluating the performance of an outbreak-detection algorithms using semi-synthetic approach: cumulative sum algorithm. *irje* 2013; 9(2): 29–38 <http://irje.tums.ac.ir/article-1-5065-en.html>
- Solgi M, Karami M, Poorolajal J. Timely detection of influenza outbreaks in Iran: evaluating the performance of the exponentially weighted moving average. *J Infect Public Health*. 2018;11(3):389–92.
- Karami M, Soori H, Mehrabi Y, Haghdooost A, Gouya M. Detecting and removing the explainable patterns of the daily counts of suspected cases of measles as a prediagnostic data source in Iran. *irje* 2012; 8(3):12–21 <http://irje.tums.ac.ir/article-1-5016-en.html>
- Martinez Beltrán ET, QuilesPérez M, Pastor-Galindo J, Nespoli P, GarciaClemente FJG. Gomez Mármol F. ConViDA: COVID-19 multidisciplinary data collection and dashboard. *J Biomed Inform*. 2021;117:103760. <https://doi.org/10.1016/j.jbi.2021.103760>
- Lubogo M, Karanja MJ, Mdodo R, Elnossery S, Osman AA, Abdi A, et al. Evaluation of the electronic early warning and Response Network (EWARN) system in Somalia, 2017–2020. *Confl Health*. 2022;16(1):1–12.
- Hosseini S, Karami M, Farhadian M, Mohammadi Y. Seasonal activity of influenza in Iran: application of influenza-like illness data from sentinel sites of healthcare centers during 2010 to 2015. *J Epidemiol Global Health*. 2018;8(1–2):29.
- Wang X, Wu S, MacIntyre CR, Zhang H, Shi W, Peng X, et al. Using an adjusted Serfling regression model to improve the early warning at the arrival of peak timing of influenza in Beijing. *PLoS ONE*. 2015;10(3):e0119923.

41. Jonnalagadda P, Swoboda C, Singh P, Gureddygar H, Scarborough S, Dunn I, et al. Developing Dashboards Address Children's Health Disparities Ohio. 2022;13(01):100–12.
42. Dixon BE, Grannis SJ, McAndrews C, Broyles AA, Mikels-Carrasco W, Wiensch A et al. Leveraging data visualization and a statewide health information exchange to support COVID-19 surveillance and response: application of public health informatics. 2021;28(7):1363–73.
43. Damalas T, Penney E, Cullen T, Dibner-Dunlap A, English C, Gomez J et al. Pima County COVID-19 vaccine solutions dashboard project: lessons learned.

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